

Ahmed Ikram | Data Engineer

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Final-year BSc Computer Science (Data Science & AI) student at the University of Dundee (predicted First, graduating June 2027). Built batch lakehouse and streaming platforms for BI and AI/ML consumers alike - including the retrieval, embedding, and feature pipelines that AI systems are trained and served on. Seeking a graduate Data Engineering role from June 2027.

EDUCATION

University of Dundee - BSc (Hons) Computer Science (Data Science and AI)

Sep 2023 - Present

- On track for a First-Class Honours degree. Expected Graduation: June 2027

High School of Dundee

Sep 2018 - June 2023

- Highers: A Mathematics, A Computing Science, A English, B Chemistry
- Computing Science classroom assistant - mentored junior pupils one-to-one during classwork.

TECHNICAL SKILLS

Languages & Stores: Python (pandas, NumPy, PyArrow, asyncio) · SQL · T-SQL · Bash · PostgreSQL · Redis · Alembic · ChromaDB

Orchestration, Scheduling & Transformation: Apache Airflow · Apache Kafka · dbt · Apache Spark / PySpark · systemd timers / cron · Pydantic · MLflow

Streaming & Warehousing: OpenLineage · Marquez · ClickHouse · Delta Lake · BigQuery · Databricks (DLT, Unity Catalog) · Azure SQL · Power BI

Data Quality & Observability: Great Expectations · Grafana · Prometheus · Azure Monitor · Cloud Monitoring · CloudWatch · StatsD · Evidently AI

Cloud & DevOps: AWS (ECS Fargate, EC2, S3, RDS, ElastiCache, WAF, IAM, CloudFront, Secrets Manager) · Azure (Databricks, Azure SQL (SQL Server), ADLS) · Google Cloud (GCS, BigQuery) · Terraform · Docker · GitHub Actions · Linux

PROJECTS

W3C Web Logs ETL Pipeline - ADLS to Power BI

Airflow · Databricks · dbt · PySpark · Azure SQL · Delta Lake · Terraform · Power BI · Marquez · Grafana · Prometheus · CI/CD

- Built a Databricks DLT medallion architecture (Bronze & Silver in DLT; gold star schema via dbt) on serverless compute: 153,380 rows with 0 rows dropped through 10 DLT expect-or-drop gates (7 bronze & 3 Silver)
- Consolidated 7 scalar GeoIP UDFs into a single struct-returning UDF which runs 3.5x faster and enriches 153,377 Silver rows to 99.99% known-country coverage.
- Modelled 16 dbt models as a Kimball star schema (SCD Type 1 dimensions; SCD Type-2 on GeoLocation via T-SQL MERGE in the Azure SQL load) on hashed natural keys, 18 compatibility macros compile PostgreSQL and T-SQL from one codebase; 121 dbt tests catch 213 duplicate rows on a 12-component incremental deduplication key.
- Orchestrated 4 Airflow DAGs with Dataset-triggered decoupling of ingest from transform; OpenLineage → Marquez records the cross-engine graph (DLT → Azure SQL → dbt) that Unity Catalog cannot see, including dbt test outcomes as quality facets.
- Backed a 7-page Power BI report (155.6K requests, 88 countries) with a freshness probe and 3 Grafana dashboards (23 panels, 8 Prometheus rules, 2 Azure Monitor alerts) including dual-dialect dbt CI and OIDC Azure CD.
- Provisioned the platform with a 2-part Terraform estate (30+ Azure resources, OIDC federation, rollback workflow), dual-dialect dbt CI and an OIDC authenticated Azure CD workflow.

WikiStream Real-Time Streaming Analytics

Python · ClickHouse · Grafana · BigQuery · Great Expectations · Terraform · Google Cloud · Pydantic · CI/CD

- Live streamed Wikipedia's EventStreams feed into self-hosted ClickHouse - 58.9M+ events in 3 days with 2,719/s real peak; Pydantic schema-on-write rejects malformed events to a dead-letter table before they hit disk.
- Guaranteed at-least-once delivery across reconnects and kills (durable cursor & deduplication ring buffer), proven with a live SIGKILL-mid-insert test (zero loss, zero duplicates).
- Cut dashboard query latency 15x at p50 (13x at p99) with rows scanned ~200x using three materialised views; held 5,655 events/sec (2x real peak) in a 10x synthetic burst with zero drops.

- Scheduled via systemd timers (cron): hourly ClickHouse → GCS → BigQuery export (exact-sum parity check every run) for a 30-day-plus warehouse tier, and hourly Great Expectations checks on freshness, nulls, and cardinality - schema drift guarded in CI.
- Itemised projected run-rate of \$41.65/month on Terraform & Workload Identity Federation GCP; 99.4% consumer-core coverage.

ATM Log Aggregation, Anomaly Detection & Diagnostics - Event Pipeline for NCR Atleos

Apache Kafka · FastAPI · PostgreSQL · Redis · ChromaDB · MLflow · Docker · Terraform · AWS · SageMaker · CI/CD

- Replaced a direct-DB generator with a distributed multi-host event pipeline: Kafka (KRaft) at-least-once delivery with idempotent deduplication (Redis SET+TTL, LRU fallback) Redis Stream DLQ with exponential backoff, ~100 msg/s over 2.5M events across 7 log sources.
- Built a Hybrid RAG Agent with LangGraph: consumers streams logs into ChromaDB through semantic chunking and local embedding - the vector store the RAG agent retrieves from over its MCP tooling.
- Stored logs in a PostgreSQL JSONB lake - a new source is a parser, not a migration; 15 indexes (6 composite) held sub-100ms queries past 2M+ rows.
- Tracked both training runs and live production anomaly detection runs in MLflow with champion-alias promotion, so ML consumers read a gated artifact, cross-checked against a live SageMaker endpoint.
- Provisioned the AWS estate in Terraform (multi-AZ VPC, ECS Fargate, RDS, S3, CloudFront, 4 least-privilege IAM roles) validated by 1,438 tests.

StockLens Feature & Retraining Pipeline

Airflow · AWS · PyTorch · MLflow · Evidently AI · PostgreSQL · Redis · Terraform · CI/CD

- Orchestrated a Weekly Airflow retraining DAG; OHLCV ingest → 17-feature build (13indicators via a Rust/PyO3 engine) → MLflow-tracked training → champion/challenger promotion that blocks any ship not beating the live champion by >2pp.
- Integrated Evidently AI drift monitoring PSI/KS/JSD on live vs training distributions, alerting to CloudWatch before drift reaches the champion - the same quality-gate pattern as the data pipelines, applied to a model artifact.
- Delivered the gated model.pt to EFS for zero-copy inference, with S3 and a PostgreSQL model registry; prediction logs retained 90 days for offline replay.

CERTIFICATES

<u>AWS Academy Graduate - Machine Learning Foundations</u>	<i>Feb 2025</i>
<u>Foundations of Azure AI: Concepts, Capabilities, and Implementation</u>	<i>Oct 2024</i>
<u>AWS Academy Graduate - Cloud Foundations</u>	<i>Mar 2024</i>

EXPERIENCE

Beaully Mini Market , Dundee, Scotland <i>Retail Assistant</i>	<i>Sep 2023 - Present</i>
Zulus Peri Peri Grill , Dundee, Scotland <i>Front of House Worker</i>	<i>June 2021 - Sep 2023</i>